

Question			Solution	Marks
2	(a)	(i)	$2mgR$	B1
		(ii)	Resultant force provides centripetal force $N_A - mg = \frac{mv_A^2}{R} \dots(1)$ $N_B + mg = \frac{mv_B^2}{R} \dots(2)$ From (a)(i): $\frac{1}{2}mv_A^2 - \frac{1}{2}mv_B^2 = mg(2R) \dots(3)$ Correct workings leading to $N_A - N_B = 6mg$	B1 B1 M1
	(b)	(i)	1. When $\theta = 0^\circ$, $g\text{-force} = 6.0 \Rightarrow N_A = 6.0 mg$ $N_A - mg = ma \Rightarrow a = 5.0 g \Rightarrow a/g = 5.0$ When $\theta = 90^\circ$, $g\text{-force} = 3.0 \Rightarrow N_A = 3.0 mg$ $N = ma \Rightarrow a = 3.0 g \Rightarrow a/g = 3.0$	M1 A0 M1 A0
			2. Using $g\text{-force} = \frac{v_A^2}{gR} + 3\cos\theta - 2$ Either at $\theta = 0^\circ$, $6 = \frac{v_A^2}{gR} + 3(1) - 2$ Or at $\theta = 180^\circ$, $0 = \frac{v_A^2}{gR} + 3(-1) - 2$ Or any other points from the graph	M1 A0
		(ii)	Shift curve downwards, or shift x-axis upwards by 1 $g\text{-force}$. Read x-intercept value. Accept $130^\circ \leq \theta \leq 135^\circ$.	A1
	(c)	(i)	Either one of the following: - For non-circular loop, carriage enters loop not from a horizontal track, thus there is no sudden transition from $g\text{-force} = 0$ to 6. - Radius of non-circular loop decreases as θ increases, thus $g\text{-force}$ does not decrease as much or as fast as the carriage travels to the top of the loop.	A1
		(ii)	Carriage loses KE due to resistive forces after going through the first loop, so it will exit the first loop with a lower speed. The second loop is lower so that the loss in GPE of the carriage as it moves from the first loop to the second loop compensates for the loss in KE.	A1

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3	(a)			The electromotive force of a source is the	